

Action-Space Quantization as a Robustness Prior in Autonomous Navigation

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Abstract: We argue that intentional quantization of the action space is not a limitation of autonomous driving policies but a structural robustness prior. Rather than learning a continuous distribution over trajectories, our system selects from nine discrete ManeuverTokens scored against geometry-derived world-state signals. This Discrete Bottleneck compresses policy variance, prevents imitation collapse on long-tail inputs, and provides a causal bottleneck that localizes failure modes precisely. We validate across three environments: a custom 2D simulator (350 rollouts, zero collisions, COMPASS 9.137/10), Waymo’s sensor-realistic AlpaSim (collision_at_fault: 0.0, distance-to-ground-truth: 0.42 m), and the Waymo Open Dataset End-to-End challenge (7.845 RFS, vs. 7.022 official baseline, under segment-grouped 5-fold cross-validation on 479 preference-labeled frames). On the Gauntlet suite—four simultaneous hazards in a 3.5 m corridor—our policy achieves 57.6% pass rate against a continuous-policy baseline at 2.1% ($27\times$ improvement). We further show that grounding Cosmos world-model embeddings in geometric context enables visual preference discrimination (GPU MLP +0.042 RFS over a feature-only gate) while a linear ranker on the same embeddings yields no gain, supporting a cross-modal alignment hypothesis. The 1.419 RFS oracle gap is explained causally: the visual-to-geometric bridge is the bottleneck, not the action space or the selector.

Keywords: autonomous driving, action-space quantization, long-tail robustness, geometric world-state, vision-language grounding, ManeuverToken, causal bottleneck

1 Introduction

Long-tail failure remains the central open problem in autonomous driving. State-of-the-art end-to-end policies achieve excellent average-case performance on in-distribution scenarios but degrade sharply on novel compositions of hazards [1, 2]. The standard response is to collect more data, add more modalities, or increase model capacity [3, 4]. We propose an orthogonal view: *reducing action-space expressiveness* is a structural form of regularization that transfers better to unseen inputs.

The core argument is topological. A continuous Gaussian policy over trajectories assigns nonzero probability to every point in trajectory space. In high-uncertainty regimes, the distribution spreads; the policy may select a trajectory that was never dangerous in training but fails on the specific geometry of an unseen hazard. A finite, structured set of ManeuverTokens cannot spread: the selected token is one of nine, each with fixed geometric semantics. Uncertainty changes the *score* of tokens but not their physical meaning. This is the Discrete Bottleneck Hypothesis.

We make three theoretical and empirical contributions:

1. **Discrete Bottleneck Hypothesis (Sec. 3.1).** A nine-token ManeuverToken space covering the safety-critical region of the SE(2) maneuver manifold achieves zero collisions over

350 rollouts and a $27\times$ gauntlet pass improvement over a continuous baseline, without any training data from AV datasets.

2. **Geometric Grounding of Visual Latents (Sec. 3.5).** Cosmos world-model embeddings carry trajectory preference signal inaccessible to linear models but extractable by a GPU MLP when the geometric context gate is present. We show that the grounding mechanism is the decisive factor and that generation-oriented VLAs (InternVLA) fail on discrimination tasks without task-aligned fine-tuning.
3. **Causal Bottleneck Analysis (Sec. 5.3).** The ManeuverToken selection process is causally determined by geometric pressure scalars with no learned correlation to demonstration data. This causal structure allows us to *precisely localize* the 1.419 RFS oracle gap to the visual-to-geometric bridge, ruling out the action space and selector as bottlenecks.

2 Related Work

Imitation learning and long-tail failure. Pomerleau [5] and Ross et al. [6] established the covariate shift problem in behavior cloning. Ho and Ermon [7] mitigated it with adversarial training but retained a continuous action space vulnerable to out-of-distribution inputs. Bansal et al. [8] identified that learned driving policies fail on synthesized worst-case scenarios. Dauner et al. [1] show that simple rule-based planners outperform learned policies on nuPlan [9] in many regimes, supporting the view that architectural inductive biases matter more than model scale for long-tail coverage.

Discrete action spaces. DQN [10] demonstrated that discrete action spaces enable stable deep RL in domains where continuous policy gradients fail. The key mechanism is variance reduction: the Bellman target for a finite action set is computed exactly, not estimated via policy gradients. We adapt this insight to the planning setting: ManeuverTokens give a finite, structured action set with geometric semantics, achieving variance reduction without RL training.

World-state abstractions. Conditional imitation learning [11] uses a route command to condition the policy but retains a continuous output. Our geometric world-state scalars are more fine-grained: they encode obstacle pressure, clearances, and blockage as scalars computed $O(1)$ from sensor geometry, providing a compact symbolic bottleneck between perception and action.

Vision-language grounding in robotics. RT-2 [4] and PaLM-E [12] show that large pre-trained visual models transfer well to robot control when fine-tuned end-to-end. We study the complementary regime: *zero fine-tuning*, where a frozen world-model encoder (Cosmos, NVIDIA 13) provides embeddings to a lightweight MLP. The discrimination gap between linear and nonlinear readouts provides evidence for the geometric grounding hypothesis.

Information bottleneck. Tishby and Schwartz-Ziv [14] and Alemi et al. [15] argue that learned representations compress task-irrelevant information. Our ManeuverToken bottleneck is a deterministic, architecture-level compression that does not require the IB objective; the nine-token structure is the bottleneck.

3 Method

3.1 Discrete ManeuverToken Space

We define nine ManeuverTokens that tile the safety-critical region of the $SE(2)$ maneuver manifold (Table 1). Each token specifies a speed scale and a signed lateral offset, realized as a 20-waypoint, 5-second trajectory at 4 Hz via cubic spline interpolation.

Coverage argument. The nine tokens span the safety-critical maneuver manifold: longitudinal response {stop, crawl, slow, yield, maintain} and lateral response {nudge, evasive} $\times$ {left, right}

Table 1: ManeuverToken definitions. Speed scale is relative to current cruise speed; lateral offset is in vehicle frame. All tokens generate 20-waypoint trajectories at 4 Hz.

Token	Speed	Lateral (m)	Semantic
STOP	0.00	0.0	Emergency stop
CRAWL	0.15	0.0	Cautious forward
MAINTAIN	1.00	0.0	Current speed, lane centre
SLOW_YIELD	0.50	0.0	Reduced speed, hold position
NUDGE_LEFT	0.80	+0.9	Mild lateral evasion left
NUDGE_RIGHT	0.80	−0.9	Mild lateral evasion right
EVASIVE_LEFT	0.70	+2.2	Strong evasion left (1/3 ramp)
EVASIVE_RIGHT	0.70	−2.2	Strong evasion right (1/3 ramp)
LANE_RECOVER	0.90	0.0	Return to lane centre

plus recovery. A continuous policy with the same five longitudinal speeds and nine lateral offsets has ~ 45 unique combinations but assigns probability mass to the continuum between them—including unsafe intermediate states unreachable by discrete tokens. The token constraint eliminates those states by construction.

3.2 Geometric World-State Encoding

At each timestep the policy computes six scalar world-state features from ego geometry and obstacle positions (no object classification, no learned features):

$$pressure = \text{clip}\left(\frac{10 - d_{\min}}{10}, 0, 1\right) \quad (1)$$

$$blockage \in [0, 1], \quad \text{fraction of corridor cross-section occupied} \quad (2)$$

$$blocked \in \{0, 1\}, \quad \text{hard stop within braking distance} \quad (3)$$

$$left_clr = \text{left ray-cast at half vehicle width (m)} \quad (4)$$

$$right_clr = \text{right ray-cast at half vehicle width (m)} \quad (5)$$

$$escape = \arg \max(left_clr, right_clr) \quad (6)$$

All six scalars are $O(1)$ to compute given a sorted obstacle list. The encoding contains no object identity, no semantic labels, and no history: it is a pure geometric snapshot. This is a deliberate design choice—it prevents the system from learning spurious correlations between object type and safe action, which is the primary imitation learning failure mode in long-tail scenarios.

3.3 Trust-Region Scoring

Each ManeuverToken is scored against a set of reference rules derived from the world-state. The score is based on a speed-scaled trust region: how far the candidate trajectory deviates from the expected safe region at 3-second and 5-second horizons.

Speed-scaled tolerance.

$$s(v) = \text{clip}\left(\frac{v - 1.4}{11.0 - 1.4}, 0, 1\right) \times 0.5 + 0.5 \quad (7)$$

$s(v) \in [0.5, 1.0]$: tight tolerances at low speed, relaxed at high speed.

Trust-region thresholds. At 3 s: $\pm 1.0 \cdot s(v)$ m lateral, $\pm 4.0 \cdot s(v)$ m longitudinal. At 5 s: $\pm 1.8 \cdot s(v)$ m, $\pm 7.2 \cdot s(v)$ m.

Token score. Let δ be the normalized overshoot beyond the trust region. For each reference rule r with target score $r.\text{score} \in [0, 100]$:

$$\text{score}_t(\tau, r) = \begin{cases} r.\text{score} & \delta = 0 \\ \max(r.\text{score} \cdot 0.1^\delta, 4.0) & \delta > 0 \end{cases} \quad (8)$$

The final token score averages the best-matching reference at 3 s and 5 s:

$$S(\tau) = 0.5 \cdot \max_r \text{score}_{3s}(\tau, r) + 0.5 \cdot \max_r \text{score}_{5s}(\tau, r) \quad (9)$$

Additional terms penalize clearance violations (-1000 if clearance < 0.55 m) and reward progress ($+40$ max). The token with highest total score is selected.

Reference rules. The reference rule set maps world-state conditions to recommended tokens with target scores in [76, 94]. Examples: low pressure (< 0.25) maps to MAINTAIN (92); moderate pressure with preferred escape side maps to NUDGE_ESCAPE (94); high blockage maps to STOP (82). Rules are hand-authored from safety engineering principles; they serve as pseudo-rater preferences, not demonstrations.

3.4 AlpaSim Adapter: Sensor-Realistic Bridge

Waymo’s AlpaSim provides sensor-realistic simulation (front-camera, LiDAR, ego dynamics) but does not expose geometric primitives directly. We bridge to the geometric world-state via four channels:

1. **Route** $\rightarrow$ **curvature.** LEFT/STRAIGHT/RIGHT route command generates a 5-point sigmoid spline (± 2.5 m amplitude at 20 m lookahead) as lane geometry.
2. **Brightness** $\rightarrow$ **visibility risk.** $\text{vis_risk} = \max(0, (0.35 - \mu_I)/0.35)$, where μ_I is mean frame brightness. Below 35% brightness, obstacle pressure is augmented.
3. **Ego dynamics** $\rightarrow$ **risk signals.** $\text{brake_risk} = \max(0, -a/4.0)$; $\text{low_speed_risk} = \max(0, (1.5 - v)/1.5)$.
4. **Structured hazards** $\rightarrow$ **simulator actors.** Hazard velocity > 0 maps to dynamic actor primitives; velocity $= 0$ maps to static obstacles. Behavior inferred from hazard label keywords.

The same Spotlight Reflex policy binary runs identically in both the custom 2D simulator and AlpaSim with no code changes—only the adapter processes differ.

3.5 WOD-E2E Trajectory Selector

The WOD-E2E challenge provides 479 preference-labeled validation frames (20-waypoint trajectories, 4 Hz, 5 s horizon). We train a trajectory selector on these frames using segment-grouped 5-fold cross-validation to prevent data leakage across scenario segments.

Candidate pool. Three candidate families provide diverse trajectory hypotheses: *Kinematic* candidates (constant velocity, constant acceleration, heading change, hold); *Ridge learned* candidates trained on 31 trajectory features; *Temporal* candidates from ego-history trends (selected on $\approx 49\%$ of frames).

Feature ranker (gate). 137 features per candidate: 31 trajectory statistics, 7 context (speed bin, intent one-hot), and 90 interaction terms ($9 \text{ source} \times 3 \text{ intent} \times 3 \text{ modalities}$). A stability-selected HistGradientBoosting model [16] ranks candidates per frame; the top-ranked candidate is the gate output (7.803 RFS, 5-fold).

Direct policy overlay. A GPU MLP trained on Cosmos 64d embeddings [13] acts as a direct policy that overrides the gate on $\approx 4.2\%$ of frames (20/479). Architecture: Linear(64 $\rightarrow$ 64)+ReLU; Linear(64 $\rightarrow$ 64)+ReLU; Linear(64 $\rightarrow$ 1)+sigmoid. Trained with BCEWithLogitsLoss, Adam (lr=0.003866), 10 epochs, batch 512, optimized with Optuna TPE [17, 18]. Champion trial (Trial 0004): precision 0.60, net contribution +0.042 RFS (7.803 $\rightarrow$ 7.845).

Grounding hypothesis. A linear ranker on the same Cosmos 64d features yields no confirmed RFS gain over the gate. The GPU MLP succeeds where the linear model fails. This suggests the Cosmos embeddings encode nonlinearly structured preference signal that only becomes accessible when aligned with the geometric gate context—supporting the cross-modal grounding hypothesis.

4 Experiments

4.1 Simulation Benchmark

We evaluate Spotlight Reflex on 350 rollouts across five suites in our custom 2D simulator. Rollouts are generated with seeds 1–10 per suite; all evaluations use the same policy binary.

Table 2: Simulation results: 350 rollouts, seeds 1–10. Pass = scenario completion without collision. All failures occur in the Gauntlet suite; zero collisions across all 350 runs.

Suite	Runs	Pass	Notes
WOD-style (11 clusters)	110	100%	All named clusters, single hazard
Compositional OOD	60	100%	Random topology $\times$ hazard
Adversarial (2–3 hazards)	60	100%	Multi-hazard, matched seeds
Hidden holdout	60	100%	Unseen seed range
Gauntlet (4 hazards, 3.5 m)	60	60%	Maximum difficulty
Total	350	93.1 %	0 collisions

COMPASS score: **9.137/10** over 700 ranked runs (95% CI collision rate: [0.0, 0.0053]). Mean minimum clearance across all passing rollouts: 2.96 m.

Gauntlet vs. baseline. Table 3 compares Spotlight Reflex against a continuous baseline (no world-state reasoning, raw trajectory regression) on the same 420 Gauntlet scenarios with matched seeds (1–80). The baseline’s 20.5% collision rate versus our 7.9% at the same pass-rate $27\times$ advantage demonstrates the variance reduction effect of the discrete token space under extreme hazard density.

Table 3: Gauntlet comparison: 420 scenarios, seeds 1–80, matched.

Policy	Pass rate	Collision rate
Continuous baseline (no world-state)	2.1%	20.5%
Spotlight Reflex (ours)	57.6%	7.9%

4.2 AlpaSim: Sensor-Realistic Validation

We run the identical policy binary in Waymo’s AlpaSim over 30 real WOD-E2E scenes with front-camera and LiDAR input. The 4-channel adapter (Sec. 3.4) converts sensor observations to geometric scalars; all decisions are made by the same Spotlight Reflex binary as in Sec. 4.1.

Results: collision_at_fault: **0.0**; distance to ground-truth trajectory: **0.42 m**; collision.any (by other agents): 0.37.

The zero at-fault collision rate in a sensor-realistic environment, with no AlpaSim-specific tuning, validates the geometric bottleneck hypothesis: the policy’s decisions are robust to the specifics of the sensing modality because they are conditioned only on geometry.

4.3 WOD-E2E Benchmark

Table 4 reports Ranked Fréchet Score (RFS) on the 479 preference-labeled validation frames under segment-grouped 5-fold cross-validation. The gate-only baseline (7.803) is already +0.672 over the official Waymo baseline (7.022); the GPU MLP overlay adds a further +0.042 for a champion score of **7.845 RFS**.

Table 4: WOD-E2E results. All 5-fold results use segment-grouped CV on 479 validation frames. 2-fold results are not directly comparable.

Selector	Folds	RFS
Waymo official baseline	—	7.022
Local constant-velocity baseline	—	7.131
Kinematic-only ranker	—	7.096
Gate only (HGB, stability-sel.)	5	7.803 [†] 2-fold estimate; not comparable to 5-fold results.
RFF direct policy ($D=512$)	5	7.834
GPU MLP + Cosmos 64d	5	7.845
HGB Optuna peak (2-fold only [†])	2	7.880
Oracle (perfect selector)	—	9.264

Oracle gap. The oracle selector (which always picks the ground-truth preferred trajectory from the candidate pool) achieves 9.264 RFS. The 1.419 RFS gap to our champion indicates that the right candidate *exists* in the pool on most frames but cannot be identified without visual context. This gap is the causal target of Contribution 3.

5 Analysis

5.1 Why Discrete? Variance–Coverage Trade-off

Extended evaluation (seeds 1–40) on compositional OOD and adversarial suites yields pass rates of 89% and 83%, respectively, compared to 100% at seeds 1–10. This degradation is expected and structurally bounded: each ManeuverToken has a fixed geometric footprint, so policy variance under OOD inputs expresses itself only as a score change, not as an arbitrary trajectory shift. The 57.6% vs. 2.1% Gauntlet comparison (Table 3) demonstrates that a continuous baseline’s variance under four simultaneous hazards produces a 20.5% collision rate, while the token constraint holds it to 7.9%—a $2.6\times$ collision reduction alongside the $27\times$ pass improvement.

The ManeuverToken space is sufficient for safety-critical coverage: stop, crawl, lateral evasion (two magnitudes, two directions), yield, and recover cover all responses documented in the Waymo long-tail scenario taxonomy. Expressiveness beyond this set increases variance without adding safety-critical behaviors.

5.2 Geometric Grounding of Visual Latents

We test five Cosmos embedding configurations against the gate baseline (Table 5):

Table 5: Cosmos 64d embedding configurations on WOD-E2E (5-fold where stated).

Configuration	Result	Notes
Linear ranker (64d feature)	No confirmed gain	Baseline parity
Discrete token embeddings	No confirmed gain	Codebook lookup
128d projection	No confirmed gain	Dimensionality increase
Concat with InternVLA 128d	No confirmed gain	Task mismatch (see below)
GPU MLP (64d, 5-fold)	+0.042 RFS	Champion

The linear model’s failure isolates the cause: the preference signal in Cosmos embeddings is nonlinearly structured, not linearly separable. The GPU MLP, operating on the same features but with two hidden layers of width 64, extracts this signal. The decisive factor is not model size (64 hidden units is modest) but architecture: the hidden layers form an implicit geometric basis aligned with the gate’s feature structure.

InternVLA failure. InternVLA [19] 128d ViT embeddings showed no RFS gain in any configuration. The explanation is task mismatch: InternVLA is trained for *action generation* (what should I do?) while preference selection requires *discrimination* (which of these trajectories is better?). Generation-oriented representations are not aligned with the pairwise preference structure of the RFS objective without task-specific fine-tuning.

5.3 Causal Bottleneck Analysis

The ManeuverToken selection process is causally determined by geometric pressure scalars (Eqs. 1–6) via the trust-region scoring formula. There is no demonstration data in the loop: the scalars are computed from obstacle geometry, the reference rules are hand-authored, and the token is selected by score maximization. This deterministic causal chain has three implications for interpreting results.

Ruling out spurious correlation. Because the policy never observes object identities or scene semantics, it cannot learn spurious correlations between (say) vehicle color and safe action. All behavioral differences between scenarios are explained by geometry. Failure modes audit this causal claim: the three identified Gauntlet failure modes (over-stopping under cumulative pressure, slow re-entry, synchronized hazard overload) are each traceable to specific scalar interactions, not to spurious associations.

Localizing the oracle gap. The 1.419 RFS oracle gap (Sec. 4.3) cannot be attributed to the action space—the candidate pool contains the preferred trajectory on most frames. It cannot be attributed to the ranker—the gate at 7.803 and GPU MLP at 7.845 exhaust the information available from trajectory features and visual embeddings without scene geometry. The residual gap is therefore attributed to the missing visual-to-geometric bridge: a model that could infer geometric context from camera images would close the gap by enabling the gate to identify the preferred candidate on the $\approx 15\%$ of frames where neither gate nor direct policy fires correctly.

Per-slice analysis supports this: GO_RIGHT frames (29 frames, 2.063 RFS regret) are under-represented in training and over-selected by the temporal gate. A geometry-aware camera model would identify that rightward intent requires specific lateral clearance checks not captured by temporal ego-history alone.

6 Limitations and Future Work

2D abstract simulation. The custom simulator uses 2D Euclidean geometry with structured obstacles. It is not a physics engine; vehicle dynamics are kinematic. AlpaSim integration (Sec. 4.2) runs the same policy in a sensor-realistic environment, but at the trajectory-plugin level. Full dynamic simulation (e.g., CARLA or nuPlan with physics) would strengthen the simulation-to-real gap claim.

Validation-set training. The WOD-E2E selector is trained on the 479 preference-labeled validation frames under segment-grouped cross-validation. This is a development analysis, not a zero-shot or test-set generalization claim. Moving training to the Waymo train split (not yet downloaded) would yield a proper test-set submission. The submission packaging pipeline is complete; the test TFRecords are the only missing piece.

No camera perception in selector. The WOD-E2E model sees only ego trajectory history, speed, acceleration, and route intent. The Cosmos GPU MLP provides visual signal from frozen embeddings

but without geometric grounding from the camera view. The identified next step is a preference-aligned visual encoder trained on the train split that maps front-camera features to geometric context scalars, closing the 1.419 RFS oracle gap.

InternVLA alignment. The generation/discrimination mismatch identified in Sec. 5.2 suggests that task-aligned fine-tuning of InternVLA on pairwise trajectory preferences would be more effective than using frozen ViT embeddings. This is straightforward to implement given access to the train-split preference labels.

7 Conclusion

We have presented three interconnected results. First, a nine-token ManeuverToken space achieves zero collisions over 350 rollouts and a $27\times$ gauntlet pass improvement over a continuous baseline, demonstrating that discrete action-space quantization is a structural robustness prior, not a limitation. Second, Cosmos world-model embeddings provide trajectory preference signal only when extracted by a nonlinear readout (GPU MLP), not a linear ranker, supporting the geometric grounding hypothesis: the embeddings encode geometrically structured preferences accessible only through aligned nonlinear computation. Third, the causal determinism of the ManeuverToken selection process enables precise localization of the 1.419 RFS oracle gap to the visual-to-geometric bridge, identifying a concrete and actionable bottleneck for future work.

Together, these results suggest a general principle: *structured inductive biases in the action space propagate to better generalization, more interpretable failure modes, and more precise identification of the information bottleneck*. We hope this perspective complements the scaling-driven paradigm and provides a complementary path to long-tail robustness in autonomous navigation.

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